Rotation of Sardinia: Palaeomagnetic Evidence from Permian Rocks

PALAEOMAGNETIC data from Middle Tertiary rocks have not decisively shown any rotation of Sardinia[1]. Rocks of Permian age, which seemed most suitable for detecting rotations of the Southern Alps[2,3], Corsica[4] and the Iberian Peninsula[5], were therefore collected in Sardinia in the hope of finding more conclusive evidence.

The samples were collected from several Permian rhyolitic ignimbrites and red sandstones in north-western Sardinia (Fig 1). The local geology is described elsewhere[6–8].

Laboratory measurements of the rock samples revealed NRM directions with south-easterly declinations and small inclinations. All samples were examined by routine demagnetization techniques with alternating fields up to 3,000 oersteds. With the ignimbrites, alternating fields

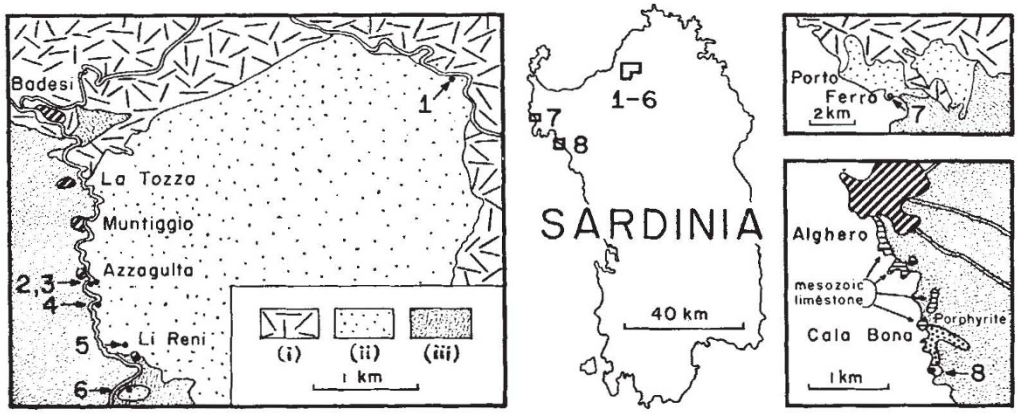


Fig. 1. Simplified geological maps of the sampling regions (after Traversa[7] and Moretti *et al.*[8]) with sampling sites. Key to hatchings: (i) rock basement below Permian volcanics; (ii) Permian volcanics and sediments; (iii) younger volcanics and sediments.

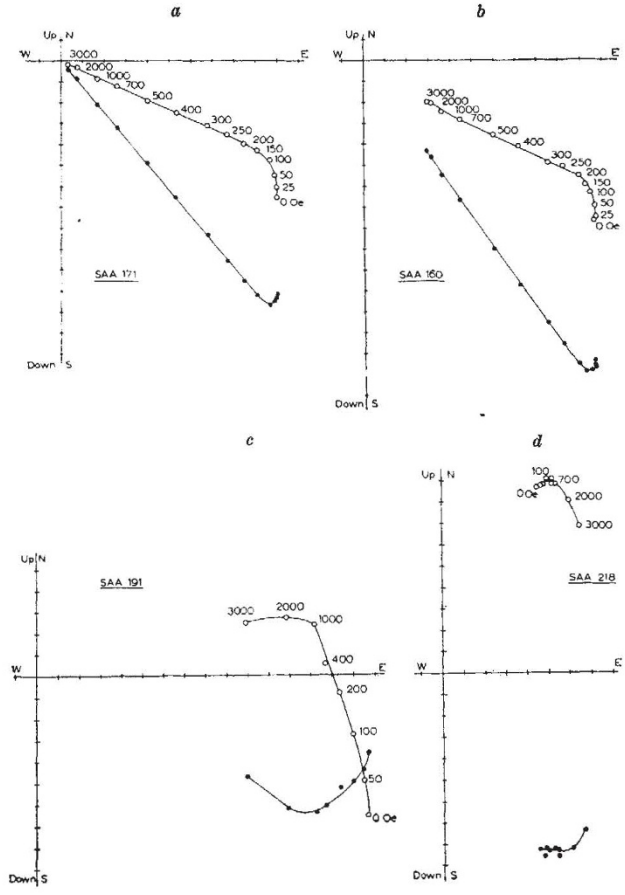


Fig. 2. Four examples of pilot sample demagnetization diagrams, from the rhyolitic ignimbrites (*a* and *b*) and the red sandstones (*c* and *d*). The points represent orthogonal projections of the successive positions of the terminus of the resultant magnetization vector during progressive demagnetization by a.c. magnetic fields. The numbers denote the intensities of these fields in oersteds. Open symbols indicate projections on the east–west vertical plane; full symbols indicate projections on the horizontal plane.

in general of less than 400 oersteds eliminated small secondary components, resulting in slight adjustments of the directions initially measured. In nearly all ignimbrite samples the total NRM could be practically eliminated at 3,000 oersteds—a typical example is given in Fig. 2*a*. Only in the samples of site 3 did an appreciable part of the NRM (with its characteristic direction) remain even after treatment at 3,000 oersteds (Fig. 2*b*).

The alternating field demagnetizations of the red sandstones of site 7 showed that their NRM is composed of a very hard characteristic component and a large and fairly hard secondary component. Although the secondary magnetizations of the sandstones could be substantially reduced with the smaller alternating fields, it seemed that a complete elimination could only be achieved with alternating fields as large as 2,000–3,000 oersteds (Fig. 2*c*). The very hard characteristic component was only slightly reduced after treatment in the large alternating fields. The NRM of the red sandstones of site 8 seems to consist only of the very hard characteristic magnetization (Fig. 2*d*). A few thermal demagnetizations of these redbeds yielded results consistent with those obtained from the alternating field demagnetizations. The dispersion of the NRM directions of the red sandstones becomes less after correction for the tectonic tilt.

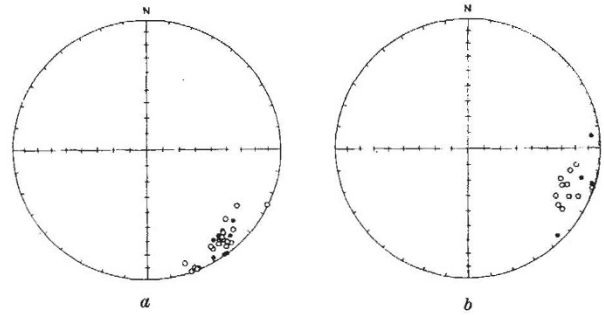


Fig. 3. Directions of magnetizations from the Permian rhyolitic ignimbrites (*a*) and the Permian red sandstones (*b*) after partial demagnetization in a.c. magnetic fields up to 3,000 oersteds and with respect to the bedding planes. Open (full) symbols denote north-seeking poles pointing up (down).

©1970 Nature Publishing Group

Table 1. SUMMARY OF PERMIAN DATA FROM SARDINIA

Site	Rock type	Bedding plane	J/cm³ × 10⁶	Q	Samples used/taken	Mean characteristic remanent magnetization direction: With respect to present horizontal D	I	K	α_{95}	With respect to bedding D	I
	Rhyolitic ignimbrites										
1		109–16	330–480	1·3–3·0	8/8	140	−1·5	74	6·5	138·5	−9·5
2		312–12	15–55	0·7–1·8	3/6	141	−8	26	24·5	142·5	−6
3		12–10	16–80	0·8–3·8	5/6	143	−4	503	3·4	143·5	−11·5
4		7–8	75–190	2·7–5·0	4/4	129·5	+10·5	27	18·0	129	+3·5
5		7 8	60–90	3·5–4·5	5/5	159	+1	454	3·6	159	−2·5
6		7–8	75–175	3·0–9·0	4/5	143	+21	733	3·4	141·5	+15·5
	Redbeds										
7		Varying	2–4	0·5–1·4	11/15	112	+15	15	12·2	112	−7·5
8		133–46	2–4	1·7–3·3	4/5	135	−34	293	5·4	108	−24
Mean direction of magnetization of the rhyolitic ignimbrites ($N = 6$)								34	11·5	142·5	−2
Mean direction of magnetization of the red sandstones ($N = 2$)								—	—	110	−15·5

Table 2. PERMIAN POLE POSITIONS

	Lat.	Long.
Stable Europe, mean	47° S	21° W
Sardinia, ignimbrites	38° S	59° E
Sardinia, redbeds	20° S	91° E
Africa, K3 Galula redbeds	46° S	40° E
Africa, K3 Ketewaka–Songwe redbeds	27° S	89° E

A comparison of the palaeomagnetic results of Sardinia with the only two Permian data available for the African continent reveals a remarkable conformity. The apparent pole position for the ignimbrites of Sardinia is situated halfway between the pole positions of the K3 Galula redbeds and the Ketewaka–Songwe redbeds of Tanganyika, and the apparent pole position for the redbeds of Sardinia is very close to the Ketewaka–Songwe pole position. Because of the restricted number of palaeomagnetic data of Sardinia as well as of Africa, however, a definite conclusion from their conformity seems to be premature.

The mean inclination values of both the Sardinian ignimbrites and the redbeds fit in well with the palaeomagnetic field pattern, extrapolated for the Sardinian region from Permian palaeomagnetic data of the European continent. The mean palaeomagnetic declination values found in the Permian rocks of Sardinia, on the other hand, deviate considerably from the declination value, extrapolated from Permian data of stable Europe. This difference is attributed to a counterclockwise rotation of Sardinia, over approximately 50° in post-Permian times, with respect to stable Europe.

This research was supported by the Netherlands Organization for the Advance of Pure Research (ZWO).

J. D. A. ZIJDERVELD

Palaeomagnetic Laboratory,
Fort Hoofddijk,
State University of Utrecht.

K. A. DE JONG
R. VAN DER VOO

Geological Institute,
State University of Utrecht.

Received March 5, 1970.

[1] De Jong, K. A., Manzoni, M., and Zijderveld, J. D. A., *Nature*, **224**, 67 (1969).
[2] Van der Voo, R., and Zijderveld, J. D. A., *Verhandelingen KNGMG*, **26**, 121 (1969).
[3] Zijderveld, J. D. A., and De Jong, K. A., *Geol. Mijnbouw*, **48**, 559 (1969).
[4] Nairn, A. E. M., and Westphal, M., *Paleogeog. Paleoclimatol. Paleoecol.*, **3**, 277 (1967).
[5] Van der Voo, R., *Tectonophysics*, **7**, 5 (1969).
[6] Vardabasso, S., *Atti Symp. Verrucano*, 293 (1966).
[7] Traversa, G., *Periodico di Mineralogia*, **35**, 513 (1966).
[8] Moretti, A., Deriu, M., Jacobacci, A., Malatesta, A., Perno, U., and Segre, A. G., *1 : 100,000 Geol. Maps 192 and 193 of Italy* (1959).

Palaeolatitudes from JOIDES Deep Sea Sediment Cores

IN the framework of plate tectonics the Earth is thought to be composed of a mosaic of rigid plates riding on top of a viscous asthenosphere. The motion of these plates with respect to the Earth's spin axis can be determined using the methods of palaeomagnetism. Measurement of continental rocks[1], oceanic islands[2] and on seamounts[3,4] all yield data which can be used to determine these motions.

Palaeomagnetic measurements of deep sea sediment cores could yield additional information about the past motions of the sea floor. Because of their short length of less than 20 m, however, these cores have been studied principally to determine the frequency and pattern of reversals in the Earth's field during the past 5–10 m.y. With the advent of the JOIDES deep sea drilling programme, cores are now available spanning most of the depositional history of the sea floor.

During Leg II of the JOIDES deep sea drilling programme[5], on station 10 (32° 52′ N, 52° 13′ W), core was recovered spanning a large enough time sequence to enable a reasonable statistical analysis to be made of the palaeomagnetic data. Twenty cores were taken in all and covered two main sequences; the first seven, the Miocene through Upper Eocene, and the last ten, almost the entire Upper Cretaceous (Campanian). Two intermediate cores sampled the Middle Eocene and the Upper Maastrichtian. Opdyke[6] has previously discussed the sequence of polarity reversals recorded in this core. Here we discuss the record of changes in inclination.

The cores on which successful palaeomagnetic measurements were made separate naturally into five groups. Though the relative declination and intensity of the samples vary considerably, the five groups after a.c. demagnetization in a field of 50 oersteds possess absolute values of inclination which are consistent within each group but different between groups. The mean value for for each group, neglecting core 5, for which the intensities were extremely low, decreases with increasing depth in the core (Table 1). The mean value for the three Miocene

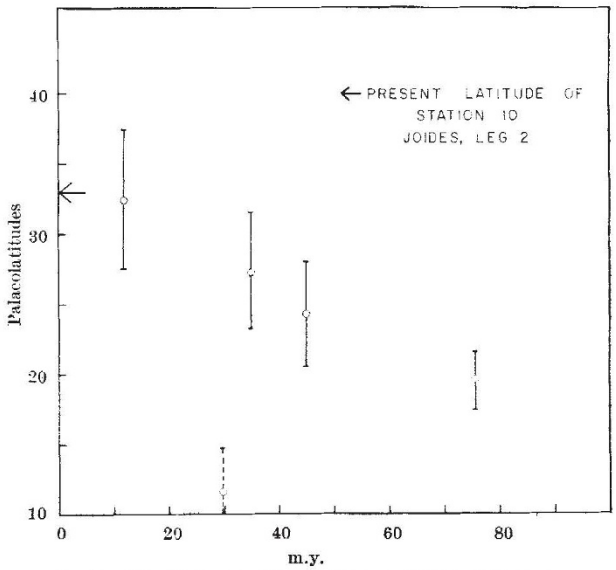


Fig. 1. Palaeolatitudes, with error bars, plotted against time.

©1970 Nature Publishing Group